

Example 12: What is the truth value of $\forall x (x^2 > x)$ if the domain consists of all real numbers? What is the truth value of the statement if the domain consists of all integers?

Soln: Let, $\exists x$ s.t. $x^2 > x$ be True. $\therefore x^2 - x > 0 \Rightarrow x(x-1) > 0$

This only holds in the following cases: ① $x > 0$ and $(x-1) > 0 \Rightarrow x > 1$ $\rightarrow$ Intersection
② $x < 0$ and $x-1 < 0 \Rightarrow x < 1$. $[x < 0] \rightarrow$ Intersection

$\therefore$ For all x , $0 < x < 1$, $x(x-1) > 0$ is False. $\therefore \forall x (x^2 > x)$ is False if the domain consists of all real numbers.

If the domain consists of all integers, $x(x-1) > 0$ is True because it holds for all $x \in \mathbb{Z}, x < 0$ and all $x \in \mathbb{Z}, x > 1$.

The Existential Quantifier

Many mathematical statements assert that there is an element with a certain property. Such statements are expressed using existential quantification. With existential quantification, we form a proposition that is true if and only if $P(x)$ is true for at least one value of x in the domain.

Definition 2: The existential quantification of $P(x)$ is the proposition "There exists an element x in the domain such that $P(x)$."

We use the notation $\exists x P(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called the existential quantifier.

A domain must always be specified when a statement $\exists x P(x)$ is used. Furthermore, the meaning of $\exists x P(x)$ changes when the domain changes. Without specifying the domain, the statement $\exists x P(x)$ has no meaning.

Example 13: Let $P(x)$ denote the statement " $x > 3$ ". What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?

Soln: 4 is a real no. $P(4): 4 > 3 \rightarrow \text{True}$. $\therefore \exists x P(x)$ is True